

Module 1A – The Universe

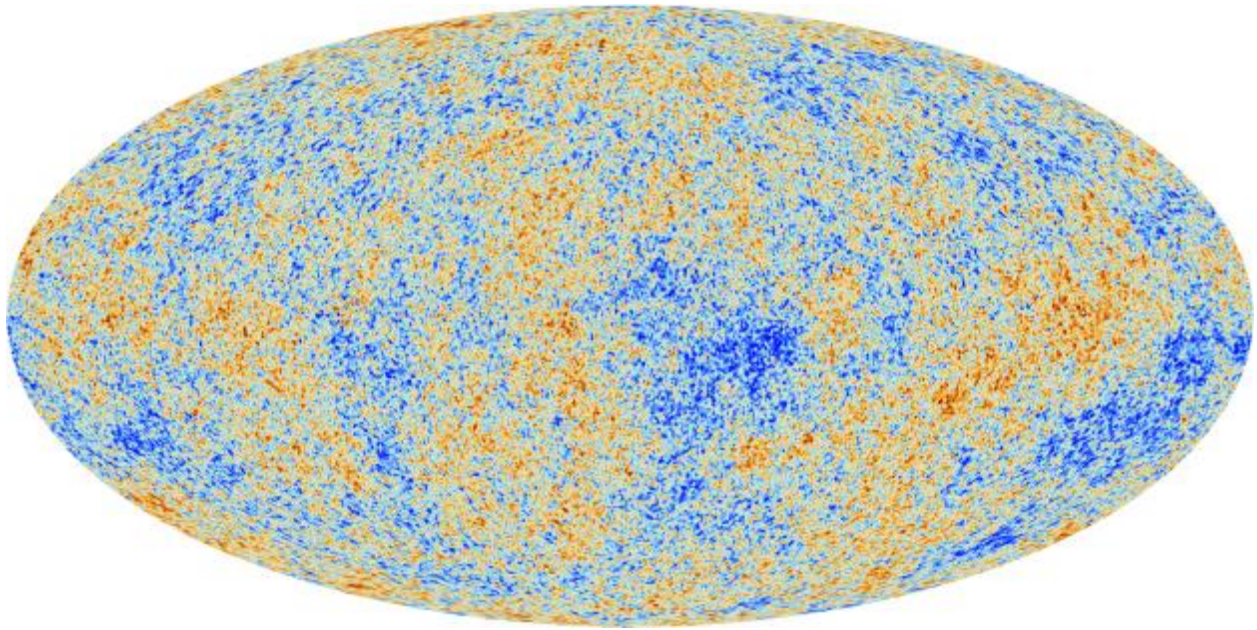


Figure 1

Astronomy is

[the scientific investigation of the Universe.](#)

The Universe is defined not as “everything,” but as

[all that exists.](#)

The critical distinction here is that “everything” is the full collection or set of things, but the Universe is more than things. For example, the Universe includes light, which is not material (a thing), and gravity waves, which are also not material (things). Additionally, the naturally occurring **laws and principles of science**, like those for gravity and electromagnetism, defined as

[the relationships between matter and force particles,](#)

are clearly not things.

One might object and say that astronomy is about “stars and stuff,” not about the Universe, and nominally that is true. The name, astronomy, comes from two Greek words, “astron” (which means stars) and “nomos” (which means arrangement of), so astronomy superficially is about the arrangements or formations of stars in the sky.

But astronomy is clearly about more than just stars. Looking up at the sky, sometimes with the aid of a telescope or binoculars, we see the Moon, the planets, comets, asteroids, meteoroids, our own Sun, and galaxies, or “star cities,” including our own galaxy, the [Milky Way](#). None of these are individual stars.

If we went to the Moon (like the Apollo astronauts), we would undoubtedly conclude that Earth, our home planet, no matter how beautiful, is just another planet circling the Sun. It follows that Earth too must be included in any full summary of astronomy.

If we then acknowledge that astronomy is the scientific investigation of the Universe, **why then do we capitalize the word Universe?**

Well the answer to this question is complex, but over the last 40 years, much has been made about how the “constants of the Universe,” like, for example,

big G, the universal gravity constant, a measure of the strength of the gravity force, and

little k, Coulomb’s constant, a measure of the strength of the electromagnetic force,

seem “rigged” in value, as if selected specifically to produce the [life we see here on Earth](#) (and presumably soon enough the life to be discovered elsewhere in the Universe).

Are these “life friendly” constants of the Universe, put in “by God” at the beginning (creation) of our Universe, at the beginning of our time, or do the laws of nature somehow require these constants to be the values that they have? Neither of these two possibilities seem to “hold water.”

It turns out though that there is third possibility, namely that there may be many universes, perhaps an infinity, of which our Universe is just one, and the constants of these other universes are not expected to work together to allow life to flourish. In such a **multiverse** (multiple universes), our Universe would be an “odd ball” universe (one out of many) that does allow life to flourish within it, and that is why we are here, now, [talking about it](#).

How ironic then that a great American poet and playwright (e.e. cummings) once [quipped](#):

“listen, there’s a hell of a good universe next door, let’s go.”

Unfortunately, we can’t go to any of these other possible universes, even if they exist.

In the meantime, though, we enjoy this Universe we live in, the one we have the privilege of existing and prospering in. Like a fish in water, we are immersed in our Universe, spelled with a capital U, never to escape.

Astronomy the subject is part of physics the subject, and the subject matter of physics is also the Universe, so what is the **distinction between astronomy and physics?**

Well, with physics, the emphasis on what is being investigated is Earth bound, whereas with astronomy, the emphasis on what is being investigated is outside of Earth. We say that

physics is mostly about what is happening here on Earth, whereas astronomy is mostly about what is happening “out there” (in outer space).

We need to make another subject matter distinction as well. **Cosmology** is technically a part of astronomy, a sort of “theology” of astronomy, that deals with ultimate science questions, like these **three major cosmological questions**:

Did the Universe have a beginning?

How did everything that exists get to be the way it is now?

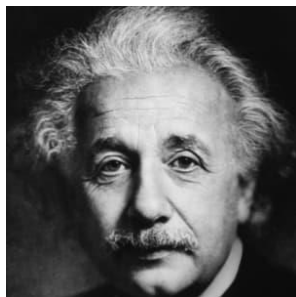
Will the Universe have an ending?

It turns out these three questions have only recently been scientifically approached and (in certain cases) definitively answered. As a result, cosmology now stands “on its own two feet” as a full-fledged and fascinating subject in **modern physics**, the physics of the twentieth century and following.

We define **cosmology** as

the study of the evolution of the Universe, past, present and future.

Notice by this definition that cosmology can supposedly predict the future of the Universe, a seriously bold statement.



This is because cosmology is predicated on a “new” theory of gravity, space and time, discovered a little over one hundred years ago by [Albert Einstein](#), perhaps the most influential physicist of all time.

This theory, which is completely factual and not a matter of speculation, is called the **General Theory of Relativity**, and it can be used to model our Universe in such a way that “the story

Figure 2

of our Universe” can be

rewound, like a movie running backwards to its beginning (if any), or

fast forwarded, to times in the future that have not yet happened,

provided substantiating observational evidence is provided as needed.



Figure 3

This is how we know (in answer to the first cosmological question) that

the Universe had a beginning 13.8 billion years ago.

Actually, two space-based studies give us complementary estimates of the age of our Universe. One, the [WMAP](#), estimated the age of the Universe to be 13.772 billion years old, while the other, the [Planck mission](#), estimated the age of our Universe to be 13.82 billion years old. Notice the amazing precision of these measurements.

Both studies involved probes that looked at the **cosmic microwave background radiation** ([CMB](#) or CMBR), which occurred about [380,000 years](#) after the beginning of the Universe (the [Big Bang](#)), when atoms and molecules first formed.

This background radiation was [discovered](#) in 1964 by two radio astronomers and engineers from Bell Telephone Labs, named [Arno Penzias and Robert Wilson](#), who were working on a microwave radio antenna for telecommunications (and other) purposes.

They kept getting annoying background “noise” which they could not attribute to any obvious manmade (or animal) cause. Ultimately they concluded that this microwave background noise was an artifact of a “long ago time” when

the Universe had cooled enough for atoms to form, and for light particles to move freely through the Universe.



Video 1

For those old enough to remember tube-like TVs (cathode ray tube TVs), when a broadcast channel was selected but no signal was being transmitted on that channel, the TV would exhibit a kind of static. That static was in fact the CMB, and is still memorialized by the [HBO channel animated logo](#). [Click the image to see this logo.]

Both the WMAP and Planck missions were able to create “falsely colored” images of the CMB (microwaves have no inherent color, as defined by humans) for the entire night sky. The “falsely colored” image of the Planck mission is shown as Figure 1 on page 1 of this module.

The reason why we’re sure that the WMAP and Planck data prove that the Universe had a beginning, is that the CMB was predicted (by Einstein’s theory) for an expanding universe.

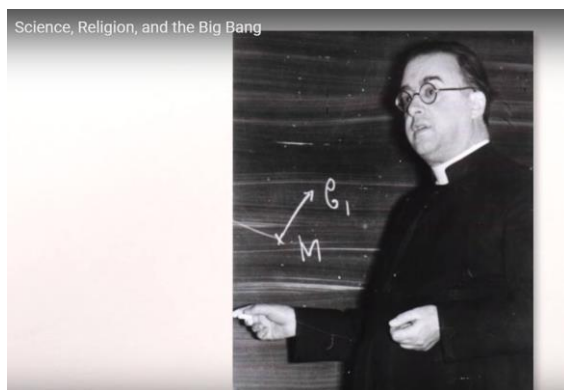
The first person to theoretically consider such a universe was a Belgian priest named [George Lemaitre](#). The first person to observe that our Universe was (and is) indeed expanding was the American astronomer [Edwin Hubble](#) (with an assist from a fellow named [Vesto Slipher](#)) who noticed that all distant galaxies were moving away from Earth, as if they were

raisins in an expanding raisin cake, all moving away from each other, with faster recessional velocities the farther away they were from each other.

The CMB, as well as other compelling evidence, forced [reluctant scientists](#) to agree that the Universe was not eternal, but had a beginning, and was much denser billions of years ago.

Explanations of the start of our Universe (the **Big Bang**) have caused much confusion. A typical follow up question is

Did the Universe start with all matter and energy at a single point? How could that possibly be?



Video 2

The current consensus is that this is not true.

Instead we say that the current visible Universe (that portion of the Universe that we can collect light from, that we can reach out to and see) was in a very small (but not zero) volume of space, with every other part of (a possibly infinite) Universe also existing in its own very small space.

Put another way,

In the beginning, the Universe was still infinite, but scaled down. Each part of it was very dense, and very hot.

[The above video – linked to Le Maitre's image -- from the [Minute Physics you tube channel](#), explains this correct way of thinking about the beginning of the Universe, in a [beautifully animated way](#).]

But why should we think that

the Universe is infinite in size?

Again to the rescue comes Mr. Einstein's theory, as well as the WMAP and Planck data. Mr. Einstein's General Theory of Relativity states that the geometry of the Universe, and correspondingly its ultimate size, is determined by how much mass and energy is in it. Put another way, the **average mass-energy density of the Universe**

ultimately determines the geometry of the Universe, and whether it is finite or infinite in extent.

Although not conclusive beyond a reasonable doubt, the WMAP and Planck data strongly suggest that the Universe

has a flat geometry (like the Euclidean geometry you learned in high school, where parallel lines never meet), and therefore, by Einstein's theory, the Universe is infinite in extent.

Since the Universe is expanding (thank you Mr. Hubble), it must be

expanding into its infinite self.

One thing we know for sure,

the Universe, if not infinite, is unimaginably large.

The “falsely colored” image of the CMB (Figure 1 of this module) reveals that the Universe was [more or less at the same temperature](#), but **not exactly at the same temperature**, when the cosmic microwave background radiation was released by newly formed atoms some 380,000 years after the Big Bang. This temperature irregularity answers our second cosmological question.

The **striking uniformity** of this CMB temperature (which was 3000° Kelvin then, but is 2.73° Kelvin now, due to the expansion of the Universe) strongly implies that the observable Universe underwent

an immense and rapid “blowing up” or “inflation” for less than a trillionth of a second after the Big Bang, with the observable Universe growing from the size of a proton to the size of a grapefruit (about 26 orders of magnitude enlargement).

During this “cosmic inflation,”

slight increases in matter density were “seeded” into the then grapefruit sized observable Universe.

The “cosmic inflation” then abruptly ended (no one is sure why), followed by the more normal “expansion of the Universe” rate we see today.

With the release of the CMB, and the subsequent formation of neutral atoms and molecules, at 380,000 years (the [recombination epoch](#)), and with the continued cooling down of the Universe (due to its relentless expansion), the **force of gravity started to dominate** the fledgling Universe, and

matter naturally started to clump together around these “seed” points of slightly higher density put into the Universe during the “cosmic inflation” period. These “seed” points grew into the stars and galaxies we see today.

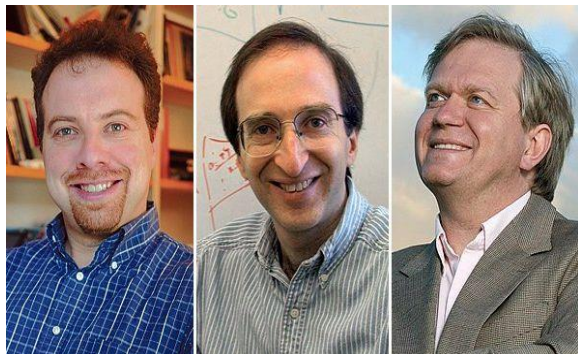


Figure 4

An additional and **monumental discovery** unrelated to the CMB was jointly published by three astronomers in 1998. Their names (left to right for the adjacent image) are Adam Riess, Saul Perlmutter, and Brian Schmidt.

They had been looking out to different distances in space for a very specific type

of exploding star called a [type 1a supernova](#).

These exploding stars have a well known **intrinsic brightness**, so it is easy for astronomers to look at them and use a little math to know exactly how far away they were when they exploded.

When looking at type 1a supernovae, they also measured something called the **cosmological redshift** for that exploding star, which [told them](#)

what the expansion rate of the Universe was at the time the star exploded.

Up until Reiss, Perlmutter and Schmidt published their paper, all astronomers thought that the Universe was dominated by the ordinary matter we see (called [baryonic matter](#), composed of protons neutrons and electrons), so it was only reasonable to think that

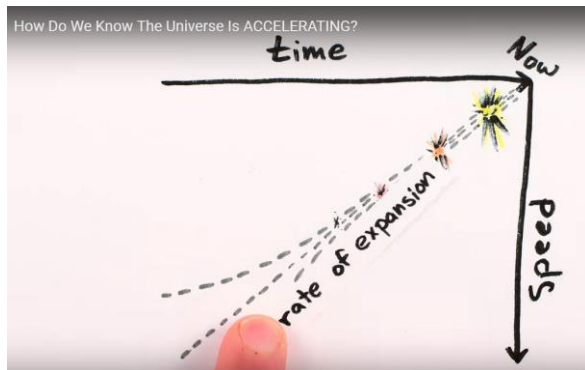
as the Universe got older, its expansion rate would slow down under the relentlessly attractive force of gravity.

Much to their (and the science world's) surprise, they discovered that the type 1a supernovae they saw were too faint for the cosmological redshift measured.

In plain English, this meant that

far from slowing down, the expansion rate of the Universe was speeding up, that the expansion rate of the Universe was accelerating.

[For more on this amazing discovery, click on the Minute Physics video [linked](#) to the image below.]



Video 3

The implications of an “accelerating Universe” are mind boggling, and bear directly on the ultimate answer to our third (and final) cosmological question:

Will the Universe have an ending?

As mentioned above, prior to 1998, when cosmologists thought the Universe was dominated by ordinary matter, it was only reasonable to conclude that the rate of expansion of the Universe would slow down. If it slowed down enough, it was thought, the expansion of the Universe could cease, and the Universe would start to contract, ultimately leading to a **Big Crunch**, where

all the mass and energy in the observable Universe would return to the form it had at the beginning of the (current) observable Universe.

With the 1998 discovery of the accelerating Universe, scientists needed to answer to **two additional cosmological questions**:

why was the Universe's expansion rate accelerating, and

what did that accelerating expansion rate mean for the future of the Universe?

The WMAP and Planck mission data offer guidance on the answer to the first additional question. We know from those missions' data that

the observable Universe is essentially flat, so it must be of a critical density to give the Universe that geometry.

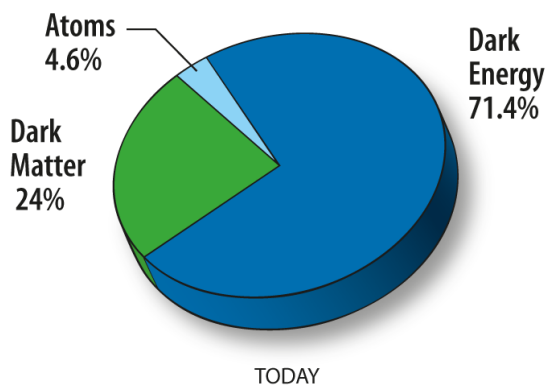


Figure 5

From Mr. Einstein's theory, we know that that critical density can only be supplied by **three kinds of matter or energy**:

ordinary matter, dark matter, and dark energy.

We define **ordinary matter** as

the regular matter we encounter every day, matter composed of protons, neutrons and electrons.

From the WMAP data, we see that ordinary matter (protons, neutrons and electrons) makes up just 4.6% of the visible Universe.

While we know a great deal about ordinary matter, by contrast we know very little about **dark matter**. We know it is matter, in the sense that it responds to the force of gravity just like ordinary matter, but it is otherwise invisible.

This is because

dark matter does not interact with light, or it is simply too faint for astronomers to see with their telescopes.

We know that dark matter must exist, because if it did not, virtually all galaxies, including our own [Milky Way](#) galaxy, would fly apart, like an unbalanced load in a washing machine in spin cycle. Clearly, galaxies do not do that: they are quite stable in their rotations.

Dark matter is thought to be one (or both) of two possible kinds: WIMPs and MACHOs (there are [other lesser possibilities](#)).

WIMPs is an acronym for “weakly interacting massive particle.” Many experiments are currently underway looking for [these particles](#).

MACHOs is an acronym for “massive astrophysical compact halo object.” They are defined as ordinary matter that is not glowing, or glowing very dimly, and so therefore can’t be seen by telescopes. An example of a [MACHO](#) would be a black hole.

The “jury is out” on whether dark matter is composed of WIMPs, or MACHOs, or a combination of the two. All we really know (from the WMAP data) is that dark matter makes up [24%](#) of the visible Universe.

So while we don’t really know what dark matter is, we know that

dark matter is more than 5 times more abundant than ordinary matter in the visible Universe.

This is a very humbling statement.

This leaves us with the third and most mysterious of the three kinds of matter and energy in our visible Universe, namely **dark energy**.

No one really knows what dark energy is, but the current leading hypothesis is that dark energy is

energy that permeates all of space and causes space to expand as if there were a repulsive gravitational force present.

Einstein’s theory allows for this dark energy in the form of a **cosmological constant** such that

the more space that is added by space expansion, the more dark energy there is.

This “cosmological constant” idea is consistent with the accelerating expansion data culled from the observations by Reiss, Perlmutter and Schmidt. Their data shows that the Universe’s expansion rate

was slowing early on in the evolution of the Universe, but then it sped up, as if dark energy’s repulsive nature overcame gravity’s attractive nature at some “crossover point” in the evolution of the Universe.

From the WMAP data, dark energy makes up 71.4% of the visible Universe.

With dark energy now ascendant in our Universe, we can hazard a guess as to the answer to our third initial, and second additional, cosmological question:

Will the Universe have an ending?

The answer seems to be that the Universe’s expansion rate will continue to accelerate, and the Universe will slowly but surely “freeze” into an oblivion of no starlight and no life.

[This oblivion](#), thought to be one hundred trillion years from now, is known figuratively as the **Big Chill**.

First Extended Study (**Advanced Students Only**):

You may have heard that in an infinite universe, any biological system (like you for instance) would be required to repeat itself (in exact form) an infinite number of times.

According to an article in phys.org,



Figure 6

If the probability of DNA-based life is greater than zero; and if the number of types of DNA-based living things is finite (because the size of the DNA life forms cannot be arbitrarily large); then an infinite universe must contain an infinite number of copies of the finite number of DNA-based living things, and some of these living things will follow very similar and even identical history lines. In other words, infinite histories plus finite types of living things means that those living things' histories are infinitely repeated.

While the “jury is still out” on this [fascinating conjecture](#), it seems clear that no one is likely to meet her “cosmic doppelganger,” because the speed of light inhibits us from visiting vast portions of the Universe.

HOMEWORK

1. Why is CMB in the microwave range of light

Normally, people think of “microwave ovens” when they hear the word microwave, but microwaves are part of a range of frequencies of light that cannot be seen by humans, but are otherwise plain old light.



Figure 7

When the CMB light was released a little less than 13.8 billion years ago, it was not in the microwave portion of light's frequencies, but in the visible portion of light's frequencies (a yellowish orange). The temperature of the Universe then was 3000°K .

The reason that the CMB “red shifted” from visible light to microwave light, is that

light wavelengths naturally “stretch” as the Universe stretches or expands.

When a light source's wavelengths increase (stretch), their frequencies decrease. Microwaves have longer wavelengths and lower frequencies than visible light.

The [current average temperature](#) of the Universe (from the CMB data) is 2.73°K .

2. Why do we associate a temperature with the CMB?



Figure 8

It's because we associate a temperature (and a color) with anything that is incandescing or glowing.

While glowing objects (sometimes called [blackbodies](#)) typically give off a continuous range of visible and invisible frequencies of light, the **intensity** of that blackbody light varies by frequency, with one frequency (and corresponding wavelength) being a

dominant frequency that you are most likely to preferentially see (if it's visible to begin with).

The dominant wavelength for blackbodies (objects that are excellent absorbers of light incident upon them) is given by [Wein's Displacement Law](#), which states that the

dominant wavelength of the blackbody is inversely proportional to the absolute temperature of the blackbody.

In plain English, this means that

the hotter a glowing body, the higher is its dominant frequency (and bluer its color).

In astronomy, we know that

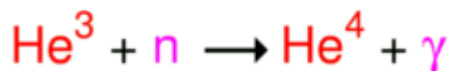
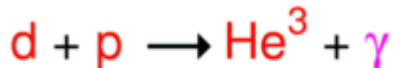
stars of the highest surface temperature glow blue, whereas stars of the lowest surface temperature glow red or even brown (which is a dull red).

The picture above is of [the Pleiades](#), a small cluster or [asterism](#) of blue (and therefore very hot) stars in the constellation Taurus.

3. Primordial Nucleosynthesis ([Advanced Students Only](#))

Most of the elements we see in [the periodic table of elements](#) were produced in the cores of active stars ([stellar nucleosynthesis](#)), but a few of them were primarily produced in the first few minutes of the Big Bang.

From about 10 seconds to about 20 minutes after the Big Bang, [the first syntheses of light atomic nuclei](#) occurred.



This process, called **primordial nucleosynthesis**, or Big Bang nucleosynthesis (BBN), produced the following atomic nuclei (atoms without accompanying electrons):

hydrogen (H), helium (He), Deuterium (D), and Lithium (Li).

Figure 9

Big Bang theory predicts that the **ratio of hydrogen (H) to Helium (He)** should be about a

3 to 1 ratio, by mass,

which is [born out](#) by WMAP and Planck Mission data.

Big Bang theory also states that the number of deuterium atoms is fixed, and were all created during primordial nucleosynthesis.

4. The Big Rip

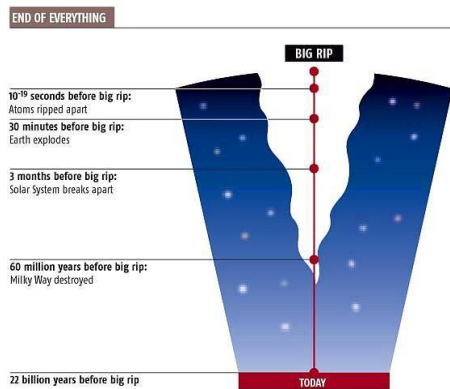


Figure 10

Besides the **Big Chill** outcome for the future of the Universe, there is [another scenario](#) being bandied about called the **Big Rip**.

In this scenario, dark energy increases dramatically over time, leading to the gradual disintegration of everything, down to the last atom. Time frame: 22 billion years from now. Likelihood based on current data: low.

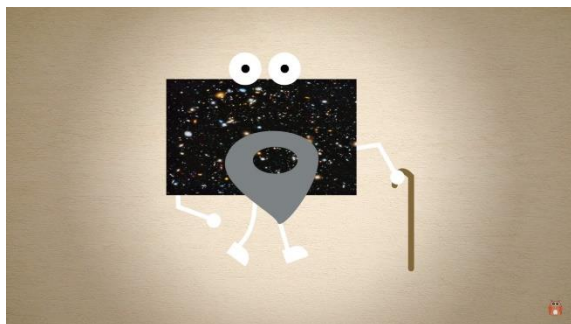
5. Olber's Paradox

Why is the sky dark at night?

Most people would answer this question with the statement “because the Sun is on the other side of the Earth, dummy!”

But many scholars through the ages, notably the German astronomer [Olbers](#), realized that this must be a legitimate question, assuming that the Universe is of infinite size and eternal (of infinite age).

The reasoning goes like this:



Video 4

In an infinite Universe of infinite age, whatever direction you look up and out at (at night), you will encounter not just the light of one star, but the light of an infinite number of stars.

The cumulative light of these stars should be blindingly bright (even though some stars would be quite distant and possible quite

dim), but even if that were not true, the whole sky should be uniformly illuminated, like our Milky Way galaxy, and this is certainly not true.

[This paradox](#), known as **Olber's paradox**, was a popular but unsolved mystery for most of the 19th century, with the American poet and short story writer [Edgar Allen Poe](#) being the first person to offer a partially correct explanation. Poe suggested that

maybe the sky is dark at night because some of the expected light has not yet arrived here at Earth.

But Poe's suggestion could only be true if the Universe was expanding and had a finite age. Since light (in the vacuum of space) has the enormous but finite speed of 186,000 miles per second, light from very distant stars could very well have not had time to reach us in the known 13.8 billion year age of Universe.

Further, in an expanding Universe that started with cosmic inflation, some starlight in an infinite Universe would never be able to get to Earth,

as more distance between those faraway stars and Earth would be added each year (due to the Universe's expansion) than the distance light could travel in that year.

One additional point must be made to fully resolve Olber's paradox:

In an expanding Universe, the earliest light (the CMB) has been redshifted from visible frequencies (orange-red) to invisible frequencies (microwaves).

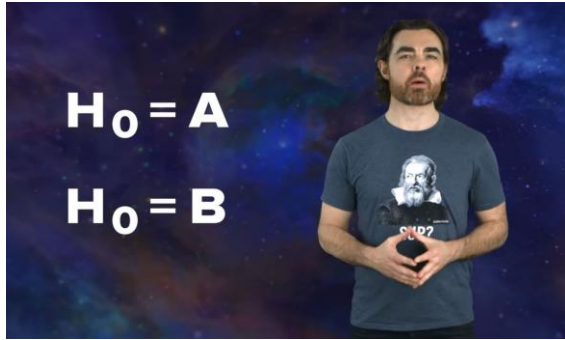
The irony of the resolution of Olber's paradox is that Olber was right:

The night sky is everywhere bathed with light (the CMB), but you just can't see it, because it is in the invisible microwave range of frequencies.

[Click](#) on the image above to see a wonderful video that thoroughly explains Olber's Paradox.

6. Hubble Trouble

The great American astronomer Edwin Hubble discovered two things of note:



Video 5

that the “smudges” in the sky called nebulae were in fact distant and distinct galaxies, and that

far away galaxies were receding from Earth at a rate proportional to how far away they were when observed.

This led to **Hubble’s Law**:

$$v_{\text{rec}} = H_0 \times d \quad (1.1)$$

where v_{rec} is the observed galaxy’s recessional velocity, in km/sec, d is the distance to the observed galaxy, and H_0 is a constant of proportionality called **Hubble’s Constant**.

For a good ninety years, determining the precise value of Hubble’s Constant has been the [holy grail](#) of astronomy.

As the [video above](#) well describes, for a long time this constant was known only within a factor of two, that is

H_0 was between 50 and 100 km/sec per Mpc,

where a megaparsec (Mpc) is 3.26 million light years distance.

But two new studies have given relatively precise measurements of H_0 .

The first study is an extension of the type 1a supernovae study of 1998, and shows that the current Hubble Constant is

$$H_0 = 73.5 \pm 1.7 \text{ km/sec per Mpc}$$

The second study, an outgrowth of the Planck CMB mission data, shows that from the time the CMB was released until now, the current Hubble Constant should have evolved to be

$$H_0 = 66.9 \pm 0.6 \text{ km/sec per Mpc}$$

The above video discusses this significant discrepancy between the two estimates of the current value of the Hubble Constant, and what reasons might contribute to this discrepancy.

LIST OF IMAGES AND VIDEOS

1. Figure 1: Image of Planck CMB map, from ESA.
2. Figure 2: Image of Albert Einstein, from Biography.com.
3. Figure 3: Image of Penzias and Wilson, from Pinterest.
4. Video 1: Video of HBO logo animation, from youtube.
5. Video 2: Video of Big Bang, from Minute Physics.
6. Figure 4: Image of Riess, Perlmutter, and Schmidt, from the New York Times.
7. Video 3: Video of Acceleration of Expansion of Universe, from Minute Physics.
8. Figure 5: Image of Planck mass-energy distribution, from Wikimedia Commons.
9. Figure 6: Image of infinite mirror regression, from Pinterest.
10. Figure 7: Image of microwave oven, from Amazon.
12. Figure 8: Image of the Pleiades, from NASA/ESA.
13. Figure 9: Image of primordial nucleosynthesis, from UCLA.
14. Figure 10: Image of the Big Rip, from NewScientist.
15. Video 4: Video on Olber's Paradox, from Smart By Design.
16. Video 5: Video on Hubble's Constant, from PBS Space Time.